Option A — Stage card

Stage 3 — Basic Engineering Gate: Critical Design Review (CDR)

The 60% design deliverables are completed and the cost and schedule are baselined. This is when the bulk of controls engineering begins.

Inputs
- Preliminary Design Package
- Developing P&IDs
- Preliminary layout

Outputs
- Preliminary I/O list
- Panel concept
- Long-lead orders placed

Controls work
- Build a preliminary I/O count and list from the P&IDs.
- Decide central vs. remote panels.
- Order long-lead items once approved.

Relevant engineering deliverables: P&ID (ready for PHA); EBOM; preliminary Interlock Matrix and Control Narrative; I/O list; Class 3 estimate.

Outgoing: Critical Design Package **Project maturity:** 20–35% **Engineering maturity:** 60%

Option B — One table per stage

Inputs	Preliminary Design Package; developing P&IDs; preliminary layout
Controls work	1. Build a preliminary I/O count and list. 2. Decide central vs. remote panels. 3. Order long-lead items once approved.
Engineering deliverables	P&ID (ready for PHA); EBOM; preliminary Interlock Matrix and Control Narrative; I/O list; Class 3 estimate
Controls outputs	Preliminary I/O list; panel concept; long-lead orders placed
Gate	Critical Design Review (CDR) → Critical Design Package; project 20–35%, engineering 60%

Option C — Input → Work → Output flow

Inputs	Controls work	Controls outputs
• Preliminary Design Package • Developing P&IDs • Preliminary layout	• Build a preliminary I/O count and list • Decide central vs. remote panels • Order long-lead items once approved	• Preliminary I/O list • Panel concept • Long-lead orders

Engineering deliverables: P&ID (ready for PHA); EBOM; preliminary Interlock Matrix and Control Narrative; I/O list; Class 3 estimate.

Gate: CDR → Critical Design Package Project 20–35% Engineering 60%